3.2.3 Composition

Similar to prior works such as Wen et al. [2024] and Li et al. [2025], we use six composition techniques to derive complete, complex instructions with the foundational *condition* and *action* components: *Single*, *And*, *Or*, *Chain*, *Selection*, *Nesting*.

3.3 Instruction Document Construction

We derive TOD-ProcBench from the ABCD dataset [Chen et al., 2021], a collection of human-written conversations created by annotators following predefined workflow actions. We aggregate all human-human conversations with the same user intent from ABCD, which inherently contain similar conversation trajectories. In Appendix B, we show examples and symbolic notations for each of the three complex instruction formats, f_1, f_2, f_3.

Format f_1 We generated "Nested If-Then" complex instructions, with one distinct instruction document for each of the 55 user intents in ABCD, using LLMs with 1-shot examples for natural language condition-action instructions (averaging 146 conversations per intent). We quantify the complexity of TOD-ProcBench with characterizations such as an average conditional branching factor of 2.2 and up to 4 nested conditional levels. To answer the question "How does the format of complex natural language instructions affect an LLM's compliance capability?", we use f_1 to derive two additional formats of the instructions which still present the fundamental *conditions* and *actions* in English natural language text.

Format f_2 "Flattened If-Then" effectively flattens the nested "If-Then" structure of f_1 into a sequence of single-level *condition-action* "If-Then" statements. We use this format to evaluate whether LLMs have a strong preference for the *Nesting* and *Chain* composition techniques present in f_1 but not f_2. The flattened structure of f_2 reduces a *Chain* into a series of composite conditions with *And*.

Format f_3 "Flattened JSON Mappings of Conditions and Actions" transforms the natural language "If-Then" statements in f_2 into a structured JSON object containing a list of maps, each of which denotes relevant *conditions* and *actions* to be considered.

3.4 Expanding to Multilingual Conversations

As a practical application of our work in constructing TOD-ProcBench and a first step to evaluate the multilingual capabilities of LLMs in complex instruction-following, we seek to answer the question "Are English instructions still effective to guide multi-turn conversations in another language?". This study will help justify whether there is a need to produce a high-quality construction of instructions in other languages for production settings where English instructions may be used in global marketplaces. We translate only the multi-turn conversations component of the original ABCD dataset to six other diverse languages: Arabic, Chinese, French, German, Hindi, Spanish.

3.5 Quality Verification

We ask human annotators to assess the quality of TOD-ProcBench's English conversation pairings with the 55 generated documents in format f_1 for the test conversations of ABCD dataset. The annotators answer the following Yes/No questions shown in the user interface in Figure 3 in Appendix A:

Accuracy: Does the generated instruction accurately reflect information in the conversation? If the answer is no, the annotator should provide which part of the instruction is incorrect.

Missing: Is there any important information from the conversation that is missing from the model-generated instruction? If the answer is yes, the annotator should specify which part from the conversation is not in the instruction.

Three annotators evaluated each conversation-instruction pair (3,012 total annotations). Annotator agreement was complete for 63% of conversations, with majority voting determining "accuracy" (82%) and "missing" (15%) assessments. The top-3 accurate annotations are for the *recover_password*, *recover_username* and *search_results* intents which share similar conversation trajectories, while the bottom-3 belong to the *shirt*, *timing* and *policy* intents which can include a broad range of queries. We only focus on the 769 high-quality pairs. Using English conversation pairings with English instructions in format f_1 as a base version of TOD-ProcBench, we expand the dataset along the two dimensions of instruction format and conversation language. For the format and multilingual expansion datasets, we leverage the LLM judge to evaluate the quality of both, with 100% correctness for the format changes and 98.3% to 99.8% for each language variation.

4 Tasks

For all experiments across three tasks, three instruction formats, and seven conversation languages, we prompt the following six LLMs to conduct chain-of-thought (CoT) [Wu et al., 2023] by first generating a step-by-step reasoning and then generate the respective output: Qwen3-14B [Yang et al., 2025], Llama3.3-70B [4], Gemma3-27B-IT [5], Claude3.5-Sonnet-V1 [6], Claude3.5-Sonnet-V2 [7], Claude3.7-Sonnet [8]. Experimental results which are not presented in tables in the following sections are presented in Appendix D for completeness.

4.1 Task 1: Instruction Retrieval and Next Action Prediction

Task 1 evaluates ToDs' ability to retrieve relevant instructions and predict appropriate next actions conditioned on conversation history. For all 30 actions (details in Appendix A, C) from the ABCD dataset, we define trigger conditions that determine when each action should be executed. We also introduce an *empty* action to handle scenarios in which the customer's goal has already been met and no further action is required from the agent.

4.1.1 Dataset

To evaluate model performance in Task 1 at different stages of the user-agent interaction, we create input data by extracting all partial conversation transcripts that end with the customer's turn in a given conversation. For example, a conversation with four customer turns leads to a list of four partial conversations, each ending with a customer turn. We then pair each partial conversation with its corresponding instruction. Overall, 6953 partial conversations are extracted from 1004 test conversations.

In order to examine the accuracy of models' predictions for the agent's next action, we process the list of workflow steps from ABCD for each partial conversation history and identify actions that are most relevant to the existing agent turns in the conversation history. The first action from the remaining workflow steps is used as the ground-truth next agent action. Examples of input data for Task1 are shown in Appendix A.

4.1.2 Results

We prompt six LLMs to generate the the next agent turn and top $k = 1$ or $k = 2$ most relevant instruction given the conversation history, list of all instructions and possible agent actions with the conditions for invoking each of these actions. In computing the top $k = 2$ accuracy, we check whether any of the two relevant instructions retrieved by LLMs is the ground-truth instruction or not. All models are executed in a few-shot in-context learning [Brown et al., 2020] paradigm where we provide three demonstrations for LLMs.

Table 1 depicts the accuracy of the models in predicting the correct instruction and agent's next action. While the larger LLMs maintain a relatively more consistent performance for $k = 1$ and $k = 2$, all LLMs recorded higher accuracy for the top-$k = 2$ scenario for both overall accuracy and only instruction retrieval. Most LLMs demonstrate a preference for format f_1 and all LLMs demonstrate more difficulty in correctly predicting the agent's next action, likely attributable to the subtle distinctions between actions and the overlapping nature of their triggering conditions. We also

[4] https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct
[5] https://huggingface.co/google/gemma-3-27b-it
[6] https://www.anthropic.com/news/claude-3-5-sonnet
[7] https://www.anthropic.com/news/3-5-models-and-computer-use
[8] https://www.anthropic.com/news/claude-3-7-sonnet

Model	Overall	Instruction Retrieval	Begin	Middle	End	Best Format
Claude3.7-Sonnet	**0.3948 / 0.4310**	**0.8267 / 0.8943**	**0.3878 / 0.4481**	**0.3834** / 0.4074	0.4161 / 0.4359	f_2 / f_1
Claude3.5-Sonnet-V2	0.3312 / 0.4182	0.7311 / 0.8765	0.3270 / 0.4214	0.3333 / **0.4126**	0.3341 / 0.4204	f_1 / f_3
Claude3.5-Sonnet-V1	0.3187 / 0.3716	0.7057 / 0.8304	0.3185 / 0.3750	0.3342 / 0.3730	0.3018 / 0.3660	f_3 / f_3
Qwen3-14B	0.3122 / 0.3279	0.7792 / 0.8419	0.1645 / 0.1668	0.2828 / 0.2880	**0.5289 / 0.5728**	f_1 / f_1
Gemma3-27B-IT	0.2221 / 0.2559	0.6138 / 0.7559	0.0770 / 0.0755	0.2144 / 0.2392	0.4113 / 0.4990	f_1 / f_2
Llama3.3-70B	0.1447 / 0.2497	0.2503 / 0.6612	0.1463 / 0.2620	0.1477 / 0.2366	0.1393 / 0.2488	f_2 / f_1

Table 1: Task 1 accuracy and best performing instruction format for top $k = 1$ and $k = 2$ instruction retrieval and next agent action prediction given English conversations. Each entry is given in the form $k = 1/k = 2$. "Overall" denotes both instruction retrieval and next action prediction as opposed to only "Instruction Retrieval". "Begin", "Middle", and "End" denote the respective 1/3 segments of the conversation. The best result per column is **bolded**.

Language	Claude3.7-Sonnet	Claude3.5-Sonnet-V2	Claude3.5-Sonnet-V1	Qwen3-14B	Gemma3-27B-IT	Llama3.3-70B
EN	**0.3959/0.3932**	0.3270/0.3312	0.3019/0.2963	0.1645/0.3122	0.0770/0.2221	0.1374/0.1358
FR	**0.4141/0.4041**	0.3615/0.3613	0.3522/0.3299	0.1540/0.2976	0.0441/0.1877	0.1188/0.1146
DE	**0.3885/0.3896**	0.3483/0.3524	0.3386/0.3283	0.1242/0.2786	0.0708/0.2088	0.1262/0.1126
ES	**0.3955/0.4005**	0.3518/0.3472	0.3282/0.3268	0.1521/0.2874	0.0658/0.1978	0.1327/0.1280
ZH	**0.3808/0.3768**	0.3212/0.3209	0.3185/0.3216	0.1521/0.2983	0.0550/0.1921	0.0851/0.0840
AR	**0.3820/0.3778**	0.3007/0.3036	0.3293/0.3206	0.1726/0.2934	0.0433/0.1585	0.0298/0.0299
HI	**0.3785/0.3882**	0.3317/0.3141	0.3386/0.3365	0.1718/0.3017	0.0701/0.1878	0.0252/0.0240

Table 2: Task 1 accuracy for predicting the correct next action and retrieving the correct top $k = 1$ instruction in the beginning 1/3 of the conversation (Begin) and for every agent response (Overall), denoted by Begin/Overall entries in the table. Results are shown across models and conversation languages (English (EN), French (FR), German (DE), Spanish (ES), Chinese (ZH), Arabic (AR), Hindi (HI)). The best result per language is **bolded** and the best result per model is underlined.

compute accuracy of the models in predicting the correct instruction and action based on relative position of the given agent response in the conversation, assigning turns to beginning, middle or end based on whether they are located in the beginning, middle, or final third of the agent turns in the conversation. According to Table 1, larger models perform consistently throughout the conversation. Although smaller models perform better at instruction retrieval and action predictions as they get more information from the conversation history, all LLMs still have more room for improvement. As shown in Appendix D, larger models tend to outperform smaller models across the conversation languages but performance of all LLMs is not severely impacted by language.

4.2 Task 2: Compliance Evaluation

The main goal of Task 2 is to benchmark LLMs' capability on scoring the compliance of a given ToD's response with respect to the dialogue history and the instruction that the agent must follow. Such capability is essential to be used as both a quality verifier during inference and a proxy for improving the development process for instruction-following ToDs. We formulate a binary classification task where the goal is to distinguish between *instruction-following* and *instruction-violating* agent responses.

Dataset For the compliance evaluation task, we only focus on agent utterances that are directly relevant to instructions. To isolate such responses from generic responses such as greeting / acknowledging customer / pure empathy, we leverage a LLM to classify agent turns as *relevant* or *irrelevant* in the 6953 partial conversations from Task 1 dataset.

Compliance Manipulation Pipeline To build our binary instruction-following / instruction-violating dataset, we use human-written ABCD responses as instruction-following examples and generate instruction-violating counterparts through a four-step pipeline (Figure 2): (1) extract relevant instruction sections, (2) manipulate the extracted instructions, (3) generate instruction-violating responses based on the manipulated instructions, (4) verify whether the generated instruction-violating response entails the original ground-truth response.

1) **Instruction Section Extraction**: We use dialogue history, instruction document, and the ground-truth instruction-following agent response as input when prompting the LLM to select a section (including its closest 'if' condition) of the instruction to which the ground-truth response is directly related. If such a relevant part of the instruction is not present, we exclude this response sample.